LESSON 7.7

Represent Polygons on the Coordinate Plane

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can draw polygons from their vertex coordinates on the coordinate plane, and use coordinates to find side lengths and solve perimeter and area problems.

WHY THIS MATTERS A polygon's vertices are given as coordinates: (-4.5, 3), (5, 3), (5, -3.75), and (-4.5, -3.75). What is its perimeter?

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
vertex *Spanish: vértice*	A corner point of a polygon, named by an ordered pair on the coordinate plane.	named by an ordered pair
polygon *Spanish: polígono*	A closed figure made of line segments; on the plane you draw it by plotting the vertices and connecting them in order.	triangle, rectangle, square
perimeter *Spanish: perímetro*	The total distance around a polygon — the sum of all its side lengths.	add every side length
distance between vertices *Spanish: distancia entre vértices*	When two vertices share an x- or y-coordinate, the distance between them is the absolute difference of the other coordinates.	(2, 4) and (5, 4) share y, so the side is $\|5 - 2\| = 3$ units.
scale *Spanish: escala*	What one grid unit stands for in the real situation.	1 unit = 1 meter
Side length *Spanish: Longitud del lado*	The distance between two vertices that are joined by a side.	found with absolute value

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

A rectangle has vertices (−2, 3), (5, 3), (5, −2) and (−2, −2). Each unit is 1 meter. Find its perimeter.

1. Top side: same y, so 2 + 5 = 7 m.
2. Right side: same x, so 3 + 2 = 5 m.
3. Add all four sides: 7 + 5 + 7 + 5 = 24.

Answer: Perimeter = 24 m

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

The shape is a rectangle, so the opposite sides match: two sides of 9.5 and two of 6.75.

1. _____
2. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- Vertices _____ and _____ share the _____ -coordinate, so that side runs _____ .
- The length of that side is _____ units because _____ .
- The perimeter is _____ units because I added _____ .
- This polygon is a _____ because _____ .

Word bank vertex · polygon · perimeter · side length · coordinate plane · absolute value · horizontal · vertical · rectangle · square · add · units

WATCH OUT!

A common mistake

Subtracting coordinates without the absolute value

LESSON 7.7

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Level 0 · Most support · every hint printed

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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1 **Circle the letter of the best answer. Show how you know.**

Vertices (-4.5, 3) and (5, 3) share y = 3. What is the distance between them?

(A) 0.5 units
(B) 1.5 units
(C) 8 units
(D) 9.5 units

STEPS TO TRY

1. Which coordinate is the same? Which differs?
2. The x-values are −4.5 and 5.
3. Distance crosses zero: 4.5 to reach 0, then 5 more.

My work — one step per line	What I did to both sides

1. **Look** *What is done to the variable?*
2. **Undo** *Do the opposite — both sides.*
3. **Simplify** *One side at a time.*
4. **Check** *Put it back in.*

2 **Circle the letter of the best answer. Show how you know.**

Vertices (5, 3) and (5, -3.75) share x = 5. What is the distance between them?

(A) 0.75 units
(B) 5 units
(C) 6.75 units
(D) 8.75 units

STEPS TO TRY

1. Which coordinate differs?
2. The y-values are 3 and −3.75.
3. 3 above zero plus 3.75 below zero.

My work — one step per line	What I did to both sides

1. **Look** *What is done to the variable?*
2. **Undo** *Do the opposite — both sides.*
3. **Simplify** *One side at a time.*
4. **Check** *Put it back in.*

3 **Circle the letter of the best answer. Show how you know.**

What is the perimeter of the rectangle with side lengths 9.5 and 6.75 units?

(A) 16.25 units

(B) 26 units

(C) 32.5 units

(D) 64.125 units

STEPS TO TRY

1. A rectangle has four sides.
2. Each length appears twice.
3. 2(9.5) + 2(6.75).

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

4 **Circle the letter of the best answer. Show how you know.**

Why are two of the terms multiplied by 2 in the perimeter calculation?

(A) Because perimeter always doubles the area

(B) Because there are two coordinates in each ordered pair

(C) It is a rounding rule

(D) A rectangle's opposite sides are equal, so each length appears on two sides

STEPS TO TRY

1. Walk the rectangle: how many 9.5 sides?
2. Two of each length.
3. The 2 counts the matching sides.

SHOW YOUR WORK

EXPLAIN YOUR THINKING

Take a side you both put under 'Slanted.' Why does subtracting coordinates give the wrong length there, and what has to be true about two vertices for subtraction to work?

SENTENCE STARTER Subtracting works when the two vertices share ___, so for ___ I cannot ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help